

An Approach to Hilbert's Sixth Problem*

Yuri Ximenes Martins[†]

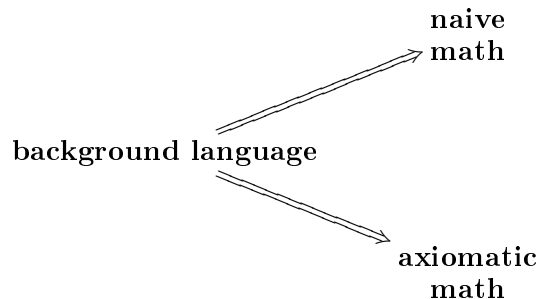
September 25, 2017

Abstract

In 1900 David Hilbert published a list with 23 problems. The sixth of these problems was on the axiomatization of the whole physics and, presently, it remains partially unsolved. In these lectures we try to convince the reader that in order to attack Hilbert's sixth problem we need to return to the foundations of mathematics and replace the usual logic by some other higher abstract language. In the first of these lectures we present Hilbert's sixth problem and discuss why usual logic is not sufficient to give a complete solution. In the second lecture, we argue that higher categorical language appears to be abstract enough to solve the problem.

1 Introduction

There are two kinds of math: *naive* (or *intuitive*) *math* and *axiomatic* (or *rigorous*) *math*. In naive math the fundamental objects are *primitives*, while in axiomatic math they are defined in terms of more elementary structures. For instance, we have *naive set theory* and *axiomatic set theory*. In both cases (naive or axiomatic) we need a *background language* (also called *logic*) in order to develop the theory, as in the diagram below. In the context of set theory this background language is just *classical logic*.



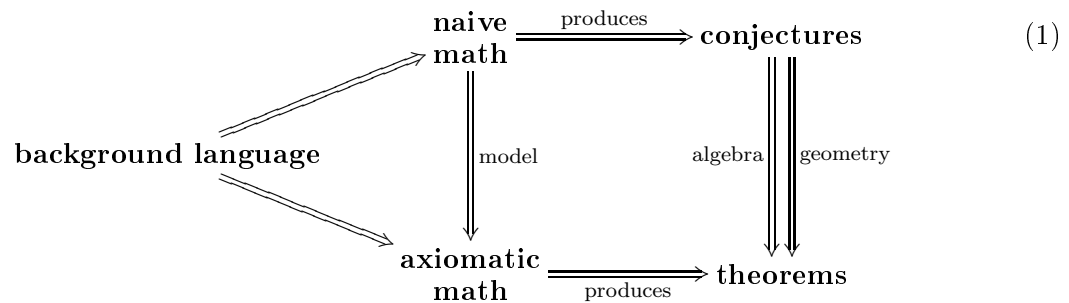
Notice that, when a mathematician is working (for instance, when he is trying to prove some new result in his area of research), at first he does not make use of completely rigorous arguments. In fact, he first uses of his intuition, doing some scribbles in pieces of paper or in a blackboard and usually considers many wrong strategies before finally discovering a good sequence of arguments which can be used to prove (or disprove) the desired result. It is only in this later moment that

*Lecture notes to a minicourse presented at the I SEMAT, 28 and 29 September 2017, held in Lavras, Brazil.

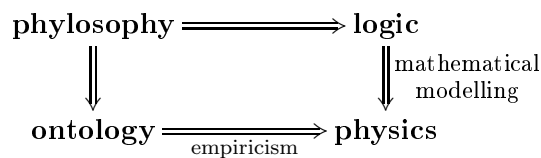
[†]Universidade Federal de Minas Gerais. Email: yurixm@ufmg.br

he tries to introduce rigor in his ideas, in order for his result to be communicated and accepted by the other members of the academy.

Thus we can say that in the process of producing new mathematics, *naive arguments come before rigorous ones*. More precisely, we can say that naive mathematics produces conjectures and rigorous mathematics turns these conjectures into theorems. It happens that the same conjecture generally can be proved (or disproved) in different ways, say by using different tools or by considering different models. For example, as will be discussed in some moments in these lectures, certain results can be proven using algebraic tools as well as geometric tools, reflecting the existence of a duality between algebra and geometry.

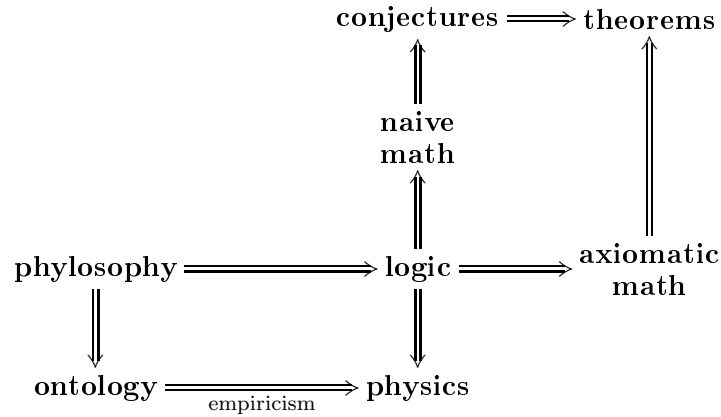


When compared with mathematics, *physics is a totally different discipline*. Indeed, in physics we have a restriction on the *existence*, meaning that we have a connection with *ontology*, which is given by *empiricism*. More precisely, *while math is a strictly logical discipline, physics is logical and ontological*. This restriction on existence produces many difficulties. For instance, logical consistence is no longer sufficient in order to establish a given sentence as physically true: ontological consistence is also needed. So, *even though a sentence is logical consistent, in order to be considered physically true it must be consistent with all possible experiments!* This fact can be expressed in terms of a commutative diagram:



But now, recall that logic determines the validity of mathematical arguments, which was also translated in terms of the commutative diagram (1). So, gluing the diagram of physics with the

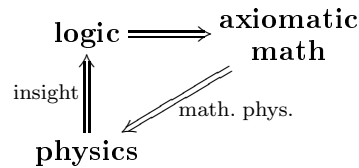
diagram of math, we have a new commutative diagram:



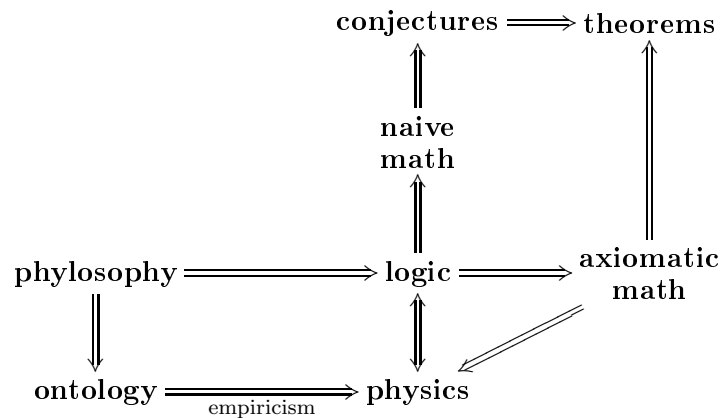
The most important fact concerning the relation between physics and mathematics is that, in the above diagram, the arrow

$$\mathbf{logic} \Longrightarrow \mathbf{physics} \quad \text{has an inverse} \quad \mathbf{physics} \Longrightarrow \mathbf{logic} .$$

This last is called *physical insight*. The composition of *physical insight* with axiomatic math produces a new arrow, called *mathematical physics*.



Therefore, the full relation between physics and mathematics is given by the following diagram



leading us to the following conclusion:

Conclusion: *we can use physical insight in order to do naive mathematics and, therefore, in order to produce conjectures. These conjectures can eventually be proven, producing theorems!*

Furthermore, with such theorems on hand we can create mathematical models for physics (mathematical physics).

Now, we can say in few words what is the fundamental objective of these lectures: *to study some examples of the following sequence*:

$$\text{physics} \Longleftrightarrow \text{logic} \Longrightarrow \begin{matrix} \text{naive} \\ \text{math} \end{matrix} \Longrightarrow \text{conjectures} \Longrightarrow \text{theorems}$$

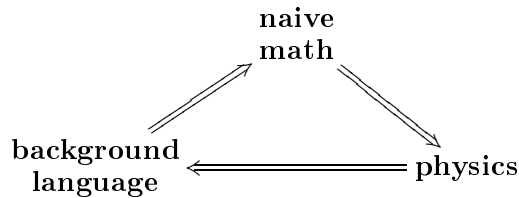
Hilbert's Sixth Problem

More honestly, our focus will be on the *axiomatization problem of physics*¹. In the year of 1900, David Hilbert published [12] a list containing 23 problems (usually known as the *Hilbert's problems*) which in his opinion would strongly influence the 20th century mathematics. The sixth of these problems is about the axiomatization of the whole physics and, presently, it remains partially unsolved. Our aim is to present an approach to this problem following works of Schreiber, Freed, Baez, Lurie and many others.

The idea is the following. Recall that the existence of physical insight gives the “mathematical physics” arrow, as below. An approach to Hilbert’s sixth problem can then be viewed as a way to present mathematical physics as a **surjective** arrow.

$$\begin{matrix} \text{axiomatic} \\ \text{math} \end{matrix} \xrightleftharpoons[\text{physics}]{\text{mathematical}} \text{physics}$$

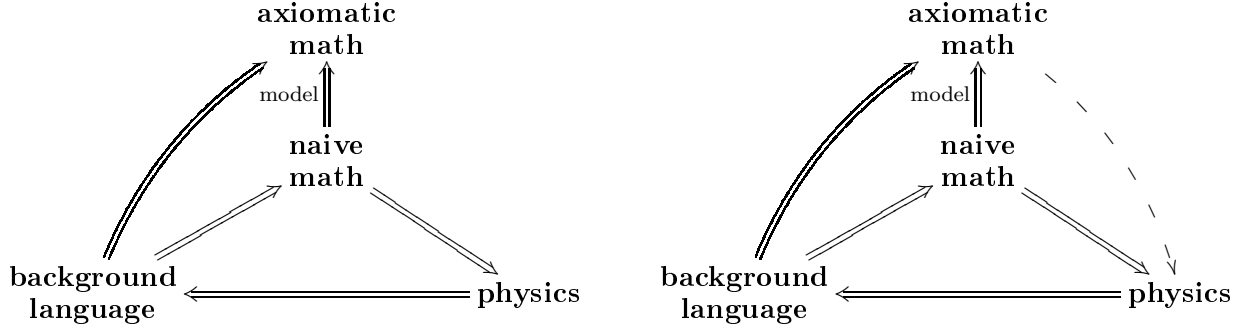
Because axiomatic math is described by some logic, the starting point is to select a proper background language. The selected background language determines directly the **naive** math, so the next step is to analyze the following loops:



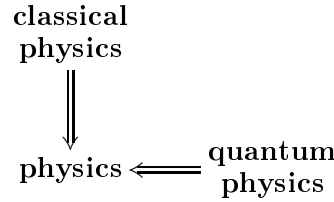
Then, selecting some model, we lift to axiomatic math, as in the first diagram below. The final step is to verify if the corresponding “mathematical physics” arrow really is surjective. In other words, *we have to verify if the axiomatic concepts produced by the selected logic are general*

¹As will be discussed, there exists another different (but strictly related) problem: the *unification of physics*.

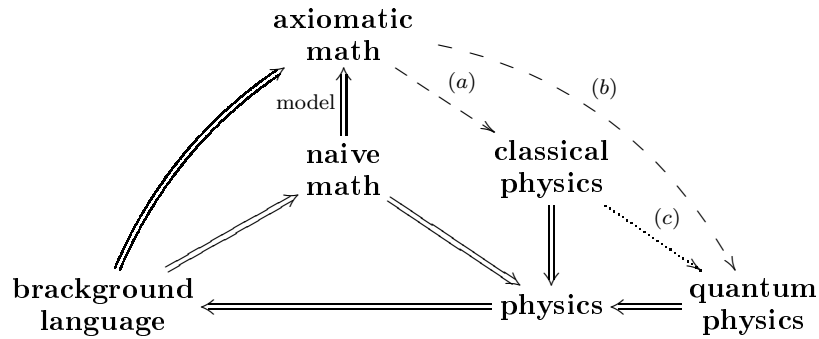
enough to model all physical phenomena.



Presently, physical theories can be divided into two classes: the *classical theories* and the *quantum theories*, as below. They are empirically classified by their domain of validity: classical theories describe those phenomena which involve the same scales of our everyday life, while quantum theories are used to describe phenomena appearing in extremal scales.



Thus, there are essentially two ways to build a surjective “mathematical physics” arrow. Either we build the arrow directly (as in the second diagram above), or we first build surjective arrows (a) and (b), which respectively axiomatize classical and quantum physics, and then another arrow (c) linking these two axiomatizations, as in the diagram below.



We emphasize the difference between the two approaches: in the first, all physical theories are described by the **same** set of axioms, while in the second classical and quantum theories are described by **different** axioms, but that are related by some process. This means that if we choose the first approach we need to *unify all physical theories*.

There are some models to this unification arrow. One of them is given by *string theory*. It is based on the assumption that the “building blocks” of nature are not particles, but indeed one dimensional entities called *strings*. More precisely, what we call “string” is just a connected one-dimensional manifold, which is diffeomorphic to some interval (when it has boundary or when it

is not compact) or to the circle (if it is boundaryless and compact). In the first case we say that we have *open strings*, while in the second we say that we have *closed strings*.

From the mathematical viewpoint, string theory is a very fruitful idea. But we need to recall that physics is not completely determined by the arrow **logic** $\Rightarrow$ **physics**. Indeed, we also have the ontological (i.e, empirical) branch. It happens that presently there is no empirical indicative that strings (instead of particles) really are the most fundamental objects of nature².

In this text, we will follow the second approach. Indeed, as we will see, if we start with a sufficiently abstract (or powerful) background language, then we can effectively axiomatize **separately** all interesting classical and quantum theories. In this context, the construction of the quantization process linking classical and quantum theories is presently incomplete. In part this comes from the fact that the underlying background language is itself under construction. Even so, there is a very promising idea, which behaves very well for many interesting cases, known as *motivic/cohomological/pull-push quantization*[19, 15].

Towards The Correct Language

As commented in the last subsection, independently of the approach used in order to attack Hilbert's sixth problem, the starting point is to select a proper background language. The most obvious choice is the *classical logic* used to describe set theory. This logic produces, via the arrow **logic** $\Rightarrow$ **axiomatic math**, *not only set theory, but indeed all classical areas of math, such as group theory, topology, differential geometry, etc!* Therefore, for this choice of background language, the different known areas of mathematics should be used in order to describe all classical and all quantum physical phenomena.

Since 1900, when Hilbert published his list of problems, many classical and quantum theories were formalized by means of these by now well known areas of mathematics. Indeed, quantum theories were observed to have more algebraic and probabilistic nature, while classical theories were presumably more geometric in character.

For instance, a system in Quantum Mechanics (which is about quantum **particles**) was formalized³ as a pair $(\mathcal{H}, \hat{H})$, where $\mathcal{H}$ is a complex (separable) Hilbert space and $\hat{H} : D(\hat{H}) \rightarrow \mathcal{H}$ is a self-adjoint operator defined in a (dense) subspace $D(\hat{H}) \subseteq \mathcal{H}$. We say that $\hat{H}$ is the *Hamiltonian* of the system and the fundamental problem is to determine its *spectrum*, which is the set of all information about $\hat{H}$ that can be accessed experimentally. The dynamics of the system from a instant t_0 to a instant t_1 is guided by the unitary operator $U(t_1; t_0) = e^{\frac{i}{\hbar}(t_1 - t_0)\hat{H}}$ operator associated to $\hat{H}$ or, equivalently, by the (time independent) Schrödinger equation $i\hbar \frac{d\psi}{dt} = \hat{H}\psi$. The probabilistic nature of Quantum Mechanics comes from the fact that the *dynamics does not determine the states of the particles, but only the time evolution of the probabilities associated to them*.

On the other hand, classical theories for **particles** are given by some *action functional* $S : X \rightarrow \mathbb{R}$, defined in some "space of fields". These fields generally involve paths $\gamma : I \rightarrow M$ in a four dimensional M , which are interpreted as the trajectories of particles moving into some "spacetime". It happens that in the typical situations, the particle is generally subjected to

²Need reference.

³Need reference.

some interaction, whose effects on the movement of the particle are measured by a corresponding quantity called *force*.

The interaction may (or not) be intrinsic to the spacetime. For instance, since the developments of General Relativity in 1916, gravity is supposed to be a intrinsic force, meaning that it will act on **any** particle. The presence of a intrinsic force is formalized by the assumption of a certain additional geometric structure on the spacetime M . For example, gravity is modeled by a Lorentzian metric g on M . Other intrinsic interactions are modeled by other types of geometric structure. It happens that **not all** manifolds may carry a given geometric structure. This means that **not all** manifolds can be used to model the spacetime. Indeed, each geometric structure exists on a given manifold only if certain quantities, called “obstruction characteristic classes” vanishes. For example, a compact manifold admits a Lorentzian metric iff its Euler characteristic $\chi(M)$ vanishes, which implies that $\mathbb{S}^4$ cannot be used to model the universe⁴.

An important class of non-intrinsic interactions are given by the Yang-Mills fields. These depend on a Lie group G , called the *gauge group*, and on a structure called a *G -principal bundle* $P \rightarrow M$ over the spacetime M . The interaction is then modelled by a connection of P , which is just a equivariant $\mathfrak{g}$ -valued 1-form $A : TP \rightarrow \mathfrak{g}$. We can think of A as the “potential” of the interaction, so that the “force” is just the (exterior covariant) derivative dA of A . Here, the standard example is electromagnetism, for which G is the abelian group $U(1)$ and A is the electromagnetic vector potential. We can think of an arbitrary Yang-Mills interaction as some kind of “nonabelian” version of electromagnetism.

Remark. There is a very important difference between classical and quantum theories. Because quantum theories are probabilistic, we can only talk about the probability of a certain event occurring in nature. Consequently, all possible configurations of a system can, in principle, be accessed. At the energy level where the classical theories lives, on the other hand, a surprising fact occurs: *only the configurations which minimize the action functional S are observed experimentally!*

Now, let us return to focus on Hilbert’s sixth problem. It requires answering for questions like these:

1. *what is a classical theory?*
2. *what is a quantum theory?*
3. *what is quantization?*

Notice that the previous discussion **does not** answer these questions. Indeed, it only reveals some examples/aspects of the classical and quantum theories; it do not say, axiomatically, what they really are. This leads us to the following conclusion:

Conclusion: *classical logic, viewed as a background language, is very nice in order to formulate and study properties of a classical or quantum theory isolately. On the other hand, it does not give tools to study more deeper questions as those required by Hilbert’s sixth problem.*

⁴For other examples of restrictions to the possible topologies of the universe, see [6, 14]

2 The Role of Higher Category Theory

The above conclusion shows that, in order to attack Hilbert's sixth problem, we need to replace classical logic by a more abstract background language. *What kind of properties this new language should have?*

Recall that by making use of classical logic we learn that classical theories are generally described by *geometric notions*, while quantum theories are described by *algebraic and probabilistic tools*. Furthermore, the quantization process should be some kind of process linking classical theories to quantum theories and, therefore, geometric areas to algebraic/probabilistic areas. So, the idea is to search for a language which formalizes the notion of “area of mathematics” and the notion of “map between two areas”.

This language actually exists: it is *categorical language*. In categorical language, an area of mathematics is determined by specifying which are the objects of interest, which are the mappings between these objects and which are the possible ways to compose two given mappings. We say that this data defines a *category*. The link between two areas of mathematics, say described by categories $\mathbf{C}$ and $\mathbf{D}$, is formalized by the notion of *functor*. This is given by a rule $F : \mathbf{C} \rightarrow \mathbf{D}$ assigning objects into objects and mappings into mappings in such a way that compositions are preserved.

Notice that we have a category **Set**, describing set theory, whose fundamental objects are sets, whose mappings are just functions between sets and whose composition laws are the usual compositions between functions. That all classical areas of mathematics can be described by categorical language comes from the fact that in each of them the fundamental objects are just *sets endowed with some further structure*, while the mappings are precisely the *functions between the underlying sets which preserve this additional structure*. For instance, Linear Algebra is the area of mathematics which study vector spaces and linear transformations. But a vector space is just a set endowed with a linear structure, while a linear transformation is just a function preserving the linear structure.

Therefore, each classical area of math defines a category $\mathbf{C}$ equipped with a “inclusion functor” $\iota : \mathbf{C} \hookrightarrow \mathbf{Set}$ which only forgets all additional structures (in the context of Linear Algebra, this functor forgets the linear structure). The categories which can be included into **Set** are called *concretes*. So, *in order to recover classical logic from categorical language it is enough to restrict to the class of concrete categories*.

The fact that categorical language is really more abstract than classical logic comes from the existence of *non-concrete* (also called *abstract*) categories. There are many of them. For instance, given a natural number p , we can build a category $\mathbf{Cob}_{p+1}$ whose objects are p -dimensional smooth manifolds and whose mappings $\Sigma : M \rightarrow N$ are *cobordisms* between them, i.e, $(p+1)$ -manifolds Σ such that $\partial\Sigma = M \sqcup N$. The abstractness of this category comes from the fact that the mappings are not functions satisfying some condition, but actually manifolds.

Example. If $p = 0$, the objects of $\mathbf{Cob}_1$ are just 0-manifolds: finite collection of points. The cobordisms between them are 1-manifolds having these 0-manifolds as boundaries. In other words, the morphisms are just disjoint unions of intervals, while the composition between intervals $[t_0; t_1]$ and $[t_1; t_2]$ is the interval $[t_0; t_2]$.

With categorical language on hand, let us try to attack Hilbert's sixth problem. We start by recalling that the dynamics of a system in Quantum Mechanics is guided by the time evolution

operators $U(t_1; t_0) = e^{\frac{i}{\hbar}(t_1 - t_0)\hat{H}}$. Notice that when varying t_0 and t_1 , all corresponding operators can be regarded as a unique functor $U : \mathbf{Cob}_1 \rightarrow \mathbf{Vec}_{\mathbb{C}}$, where $\mathbf{Vec}_{\mathbb{C}}$ denotes the category delimiting complex linear algebra (i.e, it is the category of complex vector spaces and linear transformations). Such a functor assigns to any instant t_0 a complex vector space $U(t) = \mathcal{H}_t$ and to any interval $[t_0; t_1]$ an operator $U(t_1; t_0) : \mathcal{H}_0 \rightarrow \mathcal{H}_1$.

At this point, the careful reader could do some remarks:

1. as commented previously, a system in Quantum Mechanics is defined by a pair $(\mathcal{H}, \hat{H})$, where we have **only one** space $\mathcal{H}$ which does not depend on time, so that for any interval $[t_0; t_1]$ the time evolution operator $U(t_1; t_0)$ is defined in the **same** space. On the other hand, for a functor $U : \mathbf{Cob}_1 \rightarrow \mathbf{Vec}_{\mathbb{C}}$ we have a space $\mathcal{H}_t$ for each instant of time and, therefore, for any interval the corresponding operators are defined on **different** spaces;
2. in Quantum Mechanics, the evolution is guided not by an arbitrary operator, but by a **unitary** operator. Furthermore, in Quantum Mechanics the space $\mathcal{H}$ for systems describing more than one (say k) particles decomposes as a tensor product $\mathcal{H}^1 \otimes \dots \otimes \mathcal{H}^k$. It happens that both conditions **are not** contained in the data defining a functor $U : \mathbf{Cob}_1 \rightarrow \mathbf{Vec}_{\mathbb{C}}$.

About the first remark, notice that if there is a interval $[t_0; t_1]$ connecting two different time instants t_0 and t_1 , then there is a inverse interval $[t_0; t_1]^{-1}$ such that, when composed (in $\mathbf{Cob}_1$) with the original interval, we get precisely the trivial interval. In fact, the inverse is obtained simply by flowing the time in the inverse direction. In more brief terms, any morphism in the category $\mathbf{Cob}_1$ is indeed an isomorphism. But functors preserve isomorphisms, so that for any internal $[t_0; t_1]$ the corresponding spaces $\mathcal{H}_{t_0} = U(t_0)$ and $\mathcal{H}_{t_1} = U(t_1)$ are isomorphic. This means that the spaces $\mathcal{H}_t$ **do not** dependent on the time: up to isomorphisms they are all the same.

On the second remark, let us say that there are good reasons to believe that the unitarity of the time evolution does not should be **required** as a fundamental axiom of the true fundamental physics, but instead it indeed should emerge as a consequence (or as an additional assumption) of the correct axioms. For instance, in Quantum Mechanics itself the unitarity of $U(t_0; t_1)$ can be viewed as a consequence of the Schrödinger equation, because its solutions involve the exponential of a hermitean operators which is automatically unitary⁵.

On the other hand, it is really true that an arbitrary functor $U : \mathbf{Cob}_1 \rightarrow \mathbf{Vec}_{\mathbb{C}}$ does not take into account the fact that in a system with more than one particle the total space of states decomposes as a tensor product of the state of spaces of each particle. In order to incorporate this condition, notice that when we say that we have a system of two independent particles, we are saying that time intervals corresponding to their time evolution are **disjoint**. So, we can interpret the time evolution of a system with many particles as a disconnected one dimension manifold, i.e *as a disconnected morphism of $\mathbf{Cob}_1$* . Consequently, the required condition on the space of states can be obtained by imposing the properties

$$U(t \sqcup t') \simeq U(t) \otimes U(t') \quad \text{and} \quad U(\emptyset) \simeq \mathbb{C},$$

where the second condition only means that a system with zero particles must have a trivial space of states.

⁵There are more fundamental reasons involving the possibility of topology change in quantum gravity. See [1] for an interesting discussion.

Now, notice that both categories $\mathbf{Cob}_1$ and $\mathbf{Vec}_{\mathbb{C}}$ are equipped with operations (respectively given by $\sqcup$ and $\otimes$), which are associative and commutative up to isomorphisms, together with distinguished objects (given by $\emptyset$ and $\mathbb{C}$), which behaves as “neutral elements” for these operations. A category with this kind of structure is called a *symmetric monoidal category*. A functor between monoidal categories which preserves the operations and the distinguished object is called a *monoidal functor*. Therefore, this discussion leads us to the following conclusion:

Conclusion. *A system in Quantum Mechanics is a special flavor of a monoidal functor from the category $\mathbf{Cob}_1$ of 1-dimensional cobordisms to the category $\mathbf{Vec}_{\mathbb{C}}$ of complex vector spaces.*

Consequently, with our eyes on Hilbert’s sixth problem we can use the above characterization in order to **axiomatize** a quantum theory of **particles** as being an *arbitrary monoidal functor*

$$U : (\mathbf{Cob}_1, \sqcup, \emptyset) \rightarrow (\mathbf{C}, \otimes, 1)$$

taking values in some symmetric monoidal category. A natural question is on the viability of using the same kind of argument in order to get an axiomatization of the classical theories of **particles**. This really can be done, as we will outline.

We start by recalling that a classical theory of **particles** is given by an action functional $S : X \rightarrow \mathbb{R}$, defined into some “space of fields”. Therefore, the first step is to axiomatize the notion of “space of fields”. In order to do this, recall that it is generally composed by smooth paths $\gamma : I \rightarrow M$, representing the trajectories of the particle into some spacetime M , and by the “interacting fields”, corresponding to configurations of some kind of geometric structure put in M . The canonical examples of “interacting fields” are metrics (describing gravitational interaction) and connections over bundles (describing gauge interactions).

Now we ask: *which properties smooth functions, metrics and connections have in common?* The answer is: *locality*. In fact, in order to conclude that a map $\gamma : I \rightarrow M$ is smooth, it is enough to analyze it relative to a open cover $U_i \hookrightarrow M$ given by coordinate charts $\varphi_i : U_i \rightarrow \mathbb{R}^n$. Similarly, a metric g on M is totally determined by its local components g_{ij} . Finally, it can also be shown that to give a connection $A : TP \rightarrow \mathfrak{g}$ is the same as giving a family of 1-forms $A_i : U_i \rightarrow \mathfrak{g}$ fulfilling compatibility conditions at the intersections $U_i \cap U_j$.

Therefore, we can say that a “space of fields” over a fixed spacetime M is some kind of set $\mathbf{Fields}(M)$ of structures which are “local” in the sense that, for any cover $U_i \hookrightarrow M$ by coordinate systems, we can reconstruct the total space $\mathbf{Fields}(M)$ from the subset of all $s_i \in \mathbf{Fields}(U_i)$ that are compatible in the intersections $U_i \cap U_j$.

Notice that we are searching for a notion of *space of fields* (without mention of any spacetime), but up to this point we got a notion of *space of fields over a **fixed** spacetime*. So, the main idea is to consider the rule $M \mapsto \mathbf{Fields}(M)$ and the immediate hypothesis is to suppose that it is functorial. Therefore, we can axiomatize a space of fields as a functor

$$\mathbf{Fields} : \mathbf{Diff}^{op} \rightarrow \mathbf{Set}$$

assigning to any manifold a set of local structures. This kind of functor is called a *sheaf on the site of manifolds and coordinate coverings* (or, more succinctly, a *smooth sheaf*).

This is still not the correct notion of “space of fields”. Indeed, recall that in typical situations the set $\mathbf{Fields}(M)$ contains geometric structures. It happens that when doing geometry we only

consider the entities up to their natural equivalences (their “congruences”). This means that for a fixed space M we should take the quotient space

$$\text{Fields}(M)/\text{congruences}. \quad (2)$$

In order to do calculations with a quotient space we have to **select a element** into each equivalence class and then prove that the calculus do not depend of this choice. The problem is that when we select a element we are automatically privileging it, but *there is no physically privileged element*. So, in order to be physically correct we have to work with all elements of the equivalent class **simultaneously**. This can be done by replacing the set (2) by the category whose objects are elements of $\text{Fields}(M)$ and such that there is a mapping between s, s' iff they are equivalent. In this category, all mappings are obviously isomorphisms, so that it is indeed a *groupoid*. The sheaves on the site of manifolds which take values in **Gpd** are called *smooth stacks* and they finally give the correct way of thinking about the “space of fields” for the case of **particles**.

Therefore, in order to end the axiomatization of the classical theories for **particles** we need to define what is the action functional. Given a spacetime M this should be some kind of map between the space of fields $\text{Fields}(M)$ to $\mathbb{R}$. Notice that $\mathbb{R}$ is a **set**, but we had promoted the space of fields to a **groupoid**. So, in order to define a map between these two entities we also need to promote $\mathbb{R}$ to a groupoid. This is done defining a groupoid whose objects are real numbers and having only trivial morphisms.

We then finally say that *a classical theory of **particles** is given by a smooth stack Fields (describing the space of fields) and by a rule S (describing the action functional) which assign to any manifold M a functor*

$$S_M : \text{Fields}(M) \rightarrow \mathbb{R}.$$

The Need of Highering

Up to this point we have seen that *starting with categorical logic as background language we can effectively axiomatize the notions of classical and quantum theories for **particles***. Indeed, a quantum theory is just a monoidal functor $U : \mathbf{Cob}_1 \rightarrow \mathbf{Vec}_{\mathbb{C}}$, while a classical theory is given by a smooth stack Fields and by an action functional $S_{(-)} : \text{Fields}(-) \rightarrow \mathbb{R}$.

It happens that *we do not know if particles are really the correct building blocks of nature*, so that we need a background language with allows us to axiomatize classical and quantum theories not only for particles, but for objects of any dimension. In this regard, categorical logic fails as background language. Indeed, we have at least the following problems:

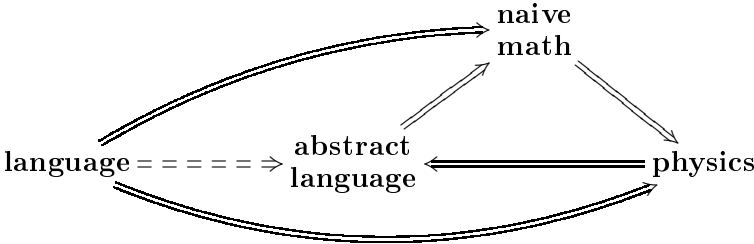
- in quantum theories. Recall that for any p we have the category $\mathbf{Cob}_{p+1}$, so that we could immediately extend the notion of quantum theory for **particles** by defining a quantum theory for **p -branes** as a monoidal functor $U : \mathbf{Cob}_{p+1} \rightarrow \mathbf{Vec}_{\mathbb{C}}$. The problem is that in this case we are only replacing the assumption that “particles are the correct building blocks of nature” by the assumption that “ p -branes are the correct building blocks of nature”. Indeed, in both cases we can only talk about quantum theories *for a specific kind of object*, while Hilbert’s sixth problem requires an *absolute notion of quantum theory*;
- in classical theories. The motion of particles (which are **0-dimensional** objects) on a spacetime M was described by smooth paths $\varphi : I \rightarrow M$, which are smooth maps defined

on a **1-dimensional** manifold I . Therefore we can easily define the motion of p -branes (which are p -dimensional objects) on M as smooth maps $\varphi : \Sigma \rightarrow M$ defined on a $(p+1)$ -dimensional manifold Σ . The problem is that in the “space of fields” of a classical theory we have to consider not only the motions but indeed the interactions which act on the object. For particles we could consider these interactions as modeled by connections because the *notion of connection is equivalent to the notion of parallel transport along paths*. Unhappily, there is no global notion of parallel transport along higher dimensional manifolds, so that *in principle the notion of connection (and, therefore, the description of the interactions) cannot be extended to the context of strings and branes*.

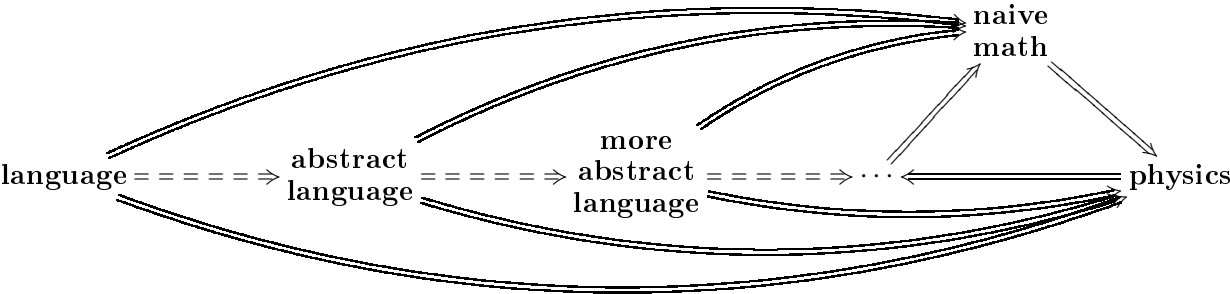
Summarizing,

Conclusion. *in order to axiomatize the notions of classical and quantum theories for higher dimensional objects we need to start with a language which is more abstract than categorical language.*

Therefore, the immediate idea is to try to build some kind of process which takes a language and returns a more abstract language. Before discussing how this can be done, recall that for a selected background language, the first step in solving Hilbert’s sixth problem is to consider a loop involving naive math, the language itself and physics. So, having constructed a more abstract language from a given one, we would like to consider loops for the new language as arising by extensions of the loop for the starting language, as in the diagram below.



This condition is very important, because it ensures that the physics axiomatized by the initial language is contained in the physics axiomatized by the new language. So, this means that when applying the process of “abstractification” we are getting languages that axiomatize more and more physics. Consequently, this seems that the iterating the process and taking the limit one hopes get a background language which is abstract enough in order to axiomatize the whole physics.



Highering

We saw that category theory is useful to axiomatize particle physics, but not string physics, so that we need to build some “abstractification process” which will be used to replace category theory by other more abstract theory. Notice that categorical language is more abstract than classical language, so that learning how to characterize the passage from classical logic to categorical logic would help to know how to iterate the construction, getting languages more abstract than categorical language. In other words, the main approach to the “abstractification process” is as some kind of “categorification process”.

In order to get some feeling on this categorification process, notice that a *set* contains less information than a *category*. Indeed, sets are composed of an unique type of information: *their elements*. On the other hand, categories have three kinds of information: *objects*, *morphisms* and *compositions*. Thus, we can understand the passage from set theory to category theory (and, therefore, from classical logic to categorical logic) as the “*addition of information layers*” (see [2] for an interesting discussion).

So, when iterating this process we expect to get a language describing entities containing more information than usual categories. Indeed, we expect to have not only objects, morphisms between objects and compositions between morphisms, but also morphisms between morphisms (called 2-morphisms) and compositions of 2-morphisms. Thus, if we call such entities *2-categories*, adding another information layer we get *3-categories*, and so on. Taking the limit we then get ∞ -*categories*, leading us to the following conclusion:

Conclusion. *A natural candidate for a background language sufficient to solve Hilbert’s sixth problem is ∞ -categorical language.*

With this conclusion in mind we need to build a math-language-physics loop to ∞ -categorical language by extending the loop obtained using categorical language. This can be done as follows:

- *in quantum theories.* Recall that the problem with the definition of quantum theories as monoidal functors $U : \mathbf{Cob}_{p+1} \rightarrow \mathbf{Vec}_{\mathbb{C}}$ involves the fact that such functors take into account only p -branes for a **fixed** p , implying that we need to know previously what is the correct building blocks of nature. Let us see that this can be avoided in the ∞ -categorical context. Indeed, we can define a ∞ -category $\mathbf{Cob}(\infty)$ having 0-manifolds as objects, 1-cobordisms (i.e cobordisms between 0-manifolds) as morphisms, 2-cobordisms (i.e, cobordisms between 1-cobordisms) as 2-morphisms and so on. Notice that differently from $\mathbf{Cob}_{p+1}$ (which contains only p -manifolds and cobordisms between them), the defined ∞ -category $\mathbf{Cob}(\infty)$ contains cobordisms of all orders and, therefore, describe p -branes for every p simultaneously. So, we can define an *absolute (or extended) quantum theory* (as required by Hilbert’s sixth problem) as some kind of ∞ -functor $U : \mathbf{Cob}(\infty) \rightarrow \infty\mathbf{Vec}_{\mathbb{C}}$, where $\infty\mathbf{Vec}_{\mathbb{C}}$ is some ∞ -categorical version of $\mathbf{Vec}_{\mathbb{C}}$ (i.e, is some ∞ -category of ∞ -vector spaces). See [3, 13];
- *in classical theories.* The problem with the axiomatization of classical theories via categorical language is that in the space of fields we have to consider interacting fields. For particles, these fields are modeled by connections on bundles, which are equivalent to parallel transport along paths. But, as commented, there is no canonical notion of transportation along higher dimensional manifolds. Another way to see the problem is the following: in order

to define a connection A locally (i.e, in terms of data over an open covering $U_i \hookrightarrow M$) we need a family of 1-forms on A_i in U_i subjected to compatibility conditions on $U_i \cap U_j$. The information U_i and $U_i \cap U_j$ belong to an usual category $\check{C}(U_i)$, whose objects are elements $x_i \in U_i$ and there is a morphism $x_i \rightarrow x_j$ iff $x_i, x_j \in U_i \cap U_j$. On the other hand, if we try to define higher transportation locally we need to take into account data on U_i which is compatible not only on $U_i \cap U_j$, but also on $U_i \cap U_j \cap U_k$, on $U_i \cap U_j \cap U_k \cap U_l$, and so on. Certainly, all this information cannot be put inside an usual category, justifying the nonexistence of connections along higher dimensional manifolds. It happens that it can be put inside a ∞ -category $\check{C}_\infty(U_i)$, meaning that we actually have a notion of ∞ -connection when we consider ∞ -categorical language (see [4, 5, 16, 17]). More precisely, the initial problem is avoided if we define the space of fields not as a smooth stack, but as a *smooth ∞ -stack*: this is a ∞ -functor $\mathbf{Fields} : \mathbf{Diff}^{op} \rightarrow \infty\mathbf{Gpd}$ such that for any M the quantity $\mathbf{Fields}(M)$ is determined not only by $\mathbf{Fields}(U_i)$ and $\mathbf{Fields}(U_i \cap U_j)$, but also by⁶

$$\mathbf{Fields}(U_i \cap U_j \cap U_k), \quad \mathbf{Fields}(U_i \cap U_j \cap U_k \cap U_l) \quad \text{and so on.}$$

Quantization

Above we concluded that ∞ -categorical language is very useful in order to attack Hilbert's sixth problem. Indeed, we saw that it gives a nice axiomatization to the notions of classical and quantum theories. But recall that in order to solve completely Hilbert's sixth problem we also need to build a quantization process linking **any** classical theory to a corresponding quantum theory. It is in this point that a complete solution fails to be given up to the present moment. Unfortunately, the question is very technical to be detailed discussed here, but we would like to explain some involved problems.

We start by noticing that with the axiomatization of classical and quantum theories on hand we can build categories $\mathbf{Class}$ and $\mathbf{Quant}$, so that it is expected that a quantization process should be some kind of functor

$$\mathcal{Q} : \mathbf{Class} \rightarrow \mathbf{Quant}.$$

Furthermore, because classical and quantum mechanics are the most simple examples of classical and quantum theories, it is natural to expect that the rule $\mathcal{Q}$ map one to another. There is a canonical way to map classical mechanics into quantum mechanics given by the so called *geometric quantization*. It happens that, by counterexamples [22, 11, 7, 8, 9] of van Hove and others, *this quantization is not functorial*.

Even so, there is a project trying to extend it to other general classical theories rather than classical mechanics. This is the so called *cohomological quantization* and it is a plain to "higher geometric quantization". In this approach we can effectively quantize any classical theory which satisfy certain "higher prequantum conditions", but the quantization itself depends of some choices and presently to determine exactly under which conditions it becomes functorial is a open problem. For a very nice review, see [20].

⁶The suggestion that not only ∞ -connections but indeed all interesting physical fields fits into a (super)smooth ∞ -stack appears is [19]

References

- [1] Baez, J. C., *Quantum Quandaries: a Category-Theoretic Perspective*, arXiv:quant-ph/0404040.
- [2] Baez, J., Dolan, J., *Categorification*, arXiv:math/9802029
- [3] Baez, J., Dolan, J., *Higher Dimensional Algebra and Topological Quantum Field Theories*, arXiv:q-alg/9503002
- [4] Baez, J. C., Schreiber, U., *Higher Gauge Theory*, arXiv:math/0511710
- [5] Baez, J., Schreiber, U., *Higher Gauge Theory: 2-Connections on 2-Bundles*, arXiv:hep-th/0412325
- [6] Dadam, F., Martins, Y. X., *Topologia, Geometria e Buracos Negros*, NEA, 2016.
- [7] Gotay, M. J., <http://arxiv.org/abs/math-ph/9809011>
- [8] Gotay, M. J., Grabowski, J., Grundling, H. B., <http://arxiv.org/abs/dg-ga/9706001>
- [9] Gotay, M. J., Grundling, H. B., <http://arxiv.org/abs/dg-ga/9710024>
- [10] Gotay, M. J., *Functorial Geometric Quantization and Van Hove's Theorem*, Int. J. Th. Ph., vol. 19, n. 2, 1980.
- [11] Groenewold, H. J., *On the Principles of elementary quantum mechanics*, Physica 12, pp. 405-460, 1946
- [12] Hilbert, D., Mathematical Problems, Bulletin of the American Mathematical Society, vol. 8, no. 10 (1902), pp. 437-479. Earlier publications (in the original German) appeared in Göttinger Nachrichten, 1900, pp. 253-297, and Archiv der Mathematik und Physik, 3dser., vol. 1 (1901), pp. 44-63, 213-237.
- [13] Lurie, J., *On the Classification of Topological Field Theories*, arXiv:0905.046.
- [14] Martins, Y. X., Biezuner, R. J., *On obstructions to the topology of the universe*, in preparation.
- [15] Nuiten, J., *Cohomological quantization of local prequantum boundary field theory*, MSc thesis, Utrecht, August 2013
- [16] Sati, H., Schreiber, U., Stasheff, J., *L-infinity algebra connections and applications to String- and Chern-Simons n-transport*, arXiv:0801.3480
- [17] Schreiber, U., Konrad, W., *Smooth Functors vs. Differential Forms*, arXiv:0802.0663
- [18] Schreiber, U., Konrad, W., *Connections on Connections on Non-Abelian Gerbes and their Holonomy*, arXiv:0808.1923
- [19] Schreiber, U., *Differential Cohomology in a Cohesive ∞ -Topos*, arXiv:1310.7930.

- [20] Schreiber, U., *What, and for what is Higher geometric quantization.*
- [21] Segal, G., *The definition of conformal field theory*, Differential geometrical methods in theoretical physics (Como, 1987), NATO Adv. Sci. Inst. Ser. C Math. Phys. Sci., 250, Kluwer Acad. Publ., Dordrecht, (1988), 165-171
- [22] van Hove, L. , Mem. Acad. Roy. de Belgique, Classe des Sci. 26 (1951) No. 6.